

Analyzing Probabilistic Models in Hierarchical BOA

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Abstract—The hierarchical Bayesian optimization algorithm (hBOA) can solve nearly decomposable and hierarchical problems of bounded difficulty in a robust and scalable manner by building and sampling probabilistic models of promising solutions. This paper analyzes probabilistic models in hBOA on four important classes of test problems: concatenated traps, random additively decomposable problems, hierarchical traps and two-dimensional Ising spin glasses with periodic boundary conditions. We argue that although the probabilistic models in hBOA can encode complex probability distributions, analyzing these models is relatively straightforward and the results of such analyses may provide practitioners with useful information about their problems. The results show that the probabilistic models in hBOA closely correspond to the structure of the underlying optimization problem, the models do not change significantly in consequent iterations of BOA, and creating adequate probabilistic models by hand is not straightforward even with complete knowledge of the optimization problem.

Index Terms—Estimation of distribution algorithms, hierarchical BOA, model structure, model complexity, probabilistic model.

I. INTRODUCTION

THE hierarchical Bayesian optimization algorithm (hBOA) [1]–[3] can solve the broad class of nearly decomposable and hierarchical problems in a robust and scalable manner. In addition, a number of efficiency enhancement techniques have been proposed to further improve hBOA performance [4]–[8]. Yet both the performance of hBOA as well as the effectiveness of various efficiency enhancement techniques for hBOA crucially depend on the quality of probabilistic models used in

hBOA to guide exploration of the space of potential solutions. Unfortunately, very little work has been done in analyzing the structure and complexity of these models with respect to the structure of the underlying optimization problem [6], [9], [10]. This knowledge is important as the more we learn about how our algorithms solve problems, the easier it is for us to develop practical theoretical models, and design new efficiency enhancements techniques and problem-specific variation operators.

The purpose of this paper is to analyze the structure and complexity of probabilistic models in hBOA on four important classes of test problems: 1) concatenated traps; 2) random additively decomposable problems; 3) hierarchical traps; and 4) two-dimensional $\pm J$ Ising spin glasses with periodic boundary conditions. Concatenated traps [11], [12] are used as an example of a separable problem of bounded order with subproblems that cannot be further decomposed [11], [13]. Random additively decomposable problems [14], [15] were chosen as an example of problems where the subproblems are of different difficulty and the amount of overlap between the different subproblems can be controlled. Hierarchical traps [1], [2] were selected as a class of difficult hierarchically decomposable problems that cannot be decomposed into independent subproblems of bounded order on a single level and that represent a challenge for most optimization methods because of their hierarchical structure, deception, and multimodality [1], [2], [16]. Just like hierarchical traps, two-dimensional Ising spin glasses [17]–[22] cannot be decomposed into subproblems of bounded order [21] and they provide a challenge for most optimization methods [18], [19], [23]–[25]; however, the structure of spin glasses is much more difficult to understand than that of hierarchical traps and it is not clear what good models should look like to solve spin glasses in a robust and scalable manner.

The results show that: 1) probabilistic models obtained by hBOA closely correspond to the structure of the underlying optimization problem; 2) the models do not change significantly between consequent iterations of hBOA; and 3) creating adequate probabilistic models by hand is far from straightforward even with complete knowledge of the structure of the underlying problem. The proposed techniques for model analysis in hBOA can be adapted to study model structure and model complexity for other problems and other estimation of distribution algorithms, providing the practitioners with a powerful tool for learning about their problems. The obtained knowledge can be used to extend the class of problems that can currently be solved tractably by hBOA through the design of new efficiency enhancement techniques and problem-specific variation operators.

This paper is organized as follows. Section II outlines hBOA. Section III discusses concatenated traps and trap-5 in particular, and then analyzes probabilistic models obtained with hBOA when solving trap-5. Section IV discusses random additively decomposable problems, and then analyzes probabilistic models

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obtained when solving various random additively decomposable problems. Section V discusses the problem of solving hierarchical traps, and then analyzes probabilistic models obtained with hBOA when solving hierarchical deceptive traps of order 3. Section VI discusses the problem of finding ground states of two-dimensional $\pm J$ Ising spin glasses with periodic boundary conditions, and presents the analysis of hBOA models in this class of problems. Section VII discusses relevant prior work. Section VIII outlines future research. Finally, Section IX summarizes and concludes this paper.

II. HIERARCHICAL BOA (hBOA)

Estimation of distribution algorithms (EDAs) [26]–[31]—also called probabilistic model-building genetic algorithms (PMBGAs) [2], [29], and iterated density estimation algorithms (IDEAs) [32]—replace standard crossover and mutation operators of genetic and evolutionary algorithms by building a probabilistic model of selected solutions and sampling the built model to generate new candidate solutions. The hBOA [1], [2], [33] is an EDA that uses Bayesian networks with local structures to represent the probabilistic model and incorporates restricted tournament replacement [34] for effective diversity maintenance. This section outlines hBOA and briefly discusses Bayesian networks, which are used to guide the exploration of the search space in hBOA and some other multivariate EDAs [35], [36].

A. Basic hBOA Procedure

hBOA evolves a population of candidate solutions represented by fixed-length strings over a finite alphabet. In this paper, we consider candidate solutions represented by binary strings of n bits.

The population is initially generated at random according to a uniform distribution over all n -bit strings. Each iteration starts by selecting a population of promising solutions using any common selection method of genetic and evolutionary algorithms, such as tournament and truncation selection. In this paper, we use truncation selection with the threshold $\tau = 50\%$, which selects the best half of the current population based on the objective function.

New solutions are generated by building a Bayesian network with decision trees [37], [38] for the selected solutions and sampling the built Bayesian network. The new candidate solutions are incorporated into the original population using restricted tournament replacement (RTR) [34]. RTR is a niching method based on crowding [39], [40], which works by localizing competition among similar solutions. It works similarly to crowding [39], but it also provides elitist pressure as the old solutions are replaced only by solutions which are at least as good as them. RTR incorporates each new candidate solution X into the original population using the following three steps: 1) Randomly select a subset W of w candidate solutions from the original population. 2) Let Y be a solution from W that is most similar to X (based on genotypic distance). 3) Replace Y with X if X is better; otherwise, discard X . We use RTR with window size $w = \min\{n, N/20\}$, where N is the size of the population, to ensure effective diversity maintenance, as suggested in [2]. The run is terminated when termination criteria are met.

B. Bayesian Networks

Bayesian networks [41], [42] combine graph theory, probability theory, and statistics to provide a flexible and practical tool for probabilistic modeling and inference. BOA and hBOA use Bayesian networks to model promising solutions found so far and sample new candidate solutions; other EDAs that use Bayesian networks to model and sample candidate solutions are the Estimation of Bayesian Networks Algorithm (EBNA) [35] and the Learning Factorized Distribution Algorithm (LFDA) [36].

A Bayesian network consists of two components: 1) *Structure*, which is defined by an acyclic directed graph with one node per variable and the edges corresponding to conditional dependencies between the variables and 2) *parameters*, which consist of the conditional probabilities of each variable given the variables that this variable depends on. Mathematically, a Bayesian network with n nodes encodes a joint probability distribution of n random variables $X_1, X_2, \dots, X_n$

$$p(X_1, X_2, \dots, X_n) = \prod_{i=1}^n p(X_i | \Pi_i) \quad (1)$$

where Π_i is the set of variables from which there exists an edge into X_i (members of Π_i are called parents of X_i) and $p(X_i | \Pi_i)$ denotes the conditional probability of X_i given Π_i . In addition to encoding direct conditional dependencies, a Bayesian network may also encode a number of conditional independence assumptions [41], [42].

hBOA uses Bayesian networks with local structures in the form of dependency trees [37], [38]. That means that the conditional probabilities for each variable are stored in a decision tree, allowing a more efficient representation of conditional dependencies and a more powerful model-building procedure. For more details on learning and sampling Bayesian networks with local structures, see [2], [37], and [38].

III. hBOA MODELS FOR SEPARABLE PROBLEMS

We start the analysis of hBOA models on separable problems of bounded difficulty; specifically, we consider concatenated traps of order 5. A separable problem is a problem in which the variables can be split into disjoint subsets and the objective function is defined as the sum of subfunctions corresponding to the different subsets of variables. A problem is said to be of bounded difficulty when it can be decomposed into subproblems of order bounded by a constant.

There are several reasons for analyzing hBOA models on these problems. First of all, many real-world problems are believed to be nearly decomposable [43], and separable problems of bounded difficulty represent a broad class of such problems [16]. Second, there is a solid body of theory that defines what a good probabilistic model should look like in order to provide scalable performance on separable problems of bounded difficulty [2], [16], [44]–[46]. Finally, concatenated traps bound the class of decomposable problems of bounded difficulty because they use fully deceptive subproblems [13], which cannot be further decomposed, and any model that fails to accurately represent the problem decomposition is expected to fail to scale up polynomially with problem size [2], [13], [16], [45], [47].

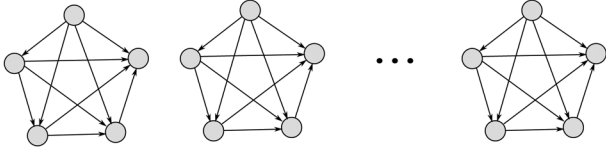


Fig. 1. A perfect model for solving trap-5 connects the variables within each trap partition but it does not contain dependencies between the different traps.

Our analysis will focus on answering two primary questions.

- 1) Do hBOA models accurately represent problem decomposition for separable problems?
- 2) How do the models change over time (from generation to generation)?

Answering the above questions is important to better understand how hBOA works on separable problems. But even more importantly, these results provide important information for developing theory and efficiency enhancement techniques for hBOA and nearly decomposable problems of bounded difficulty. Our primary focus will be on the second question as an in-depth analysis of model accuracy in hBOA on traps can be found in [10].

A. Concatenated 5-Bit Trap

In concatenated 5-bit traps [11], [13], the input string is partitioned into disjoint groups of 5 bits each. This partitioning is unknown to the algorithm, and it does not change during the run. A 5-bit trap function is then applied to each of the groups and the contributions of all the groups are added together to form the fitness, which we want to maximize. The 5-bit trap function is defined as follows:

$$\text{trap}_5(u) = \begin{cases} 5, & \text{if } u = 5 \\ 4 - u, & \text{otherwise} \end{cases} \quad (2)$$

where u is the number of ones in the input string of 5 bits. An n -bit trap-5 has one global optimum in the string of all ones and $2^{n/5} - 1$ other local optima (the number of local optima grows exponentially fast with problem size).

Trap-5 necessitates that all bits in each trap partition (linkage group) are treated together, because statistics of lower order mislead the search away from the global optimum [13], leading to highly inefficient performance [16], [45], [47].

B. A Perfect Model for Trap-5

Since the 5-bit trap described above is fully deceptive [13], to solve concatenated traps in polynomial time, it is necessary that the probabilistic model encodes conditional dependencies between all or nearly all pairs of variables within each trap partition [16], [45], [47]. On the other hand, to maximize the mixing and minimize model complexity, it is desirable that we do not discover any dependencies between the different trap partitions [16], [45], [47]. Therefore, a “perfect” model for solving trap-5 would contain all dependencies between the bits within each trap partition but it would not contain any dependencies between the different trap partitions (see Fig. 1).

In order to be able to talk about the accuracy of a model, we must define what dependencies should be included in the model

and what dependencies should be avoided. For some problems, the distinction is straightforward. For example, as mentioned above, for trap-5, the probabilistic model should include dependencies between pairs of bits in any trap partition [16], [45], [47]; on the other hand, to maximize mixing [16], [45], [47], there should be no or few dependencies between bits in different trap partitions. We call the dependencies that should be included in the model *necessary dependencies* and the dependencies that should be excluded *unnecessary dependencies*.

As is discussed later, in some problems, it is not as straightforward to distinguish the necessary and unnecessary dependencies, because the importance of various interactions may change during the run and because some of the interactions may be excluded without sacrificing reliability. Nonetheless, even for such problems, it is often possible to relate the structure of the probabilistic model to the structure of the underlying problem, as is shown in the remainder of this paper for random additively decomposable problems, hierarchical traps, and two-dimensional spin glasses.

Theory on the sufficient conditions for the interactions covered by the probabilistic model can be found in [46]. Nonetheless, as is also shown later in this paper, these sufficient conditions may sometimes yield infeasible models despite that the problem may still be solvable in low-order polynomial time. To distinguish between the dependencies between bits that exhibit epistasis, the concept of *benign* and *malign* dependencies may be used [48], [49].

C. Experimental Setup

To look at the relationship between problem size and the probabilistic models generated by hBOA, we consider traps of sizes $n = 15$ to $n = 210$ bits.

To make sure that our results correspond to the actual hBOA performance on trap-5 and are not biased by either a too large or too small population size, we first use bisection [2], [50] to ensure that the population size is large enough to obtain reliable convergence to the global optimum in 30 out of 30 independent runs. The results are then averaged over the 30 successful runs with the population size obtained by bisection. The number of generations is upper bounded according to preliminary experiments and hBOA scalability theory [44], [51] by $2\sqrt{n}$, where n is the number of bits in the problem. Each run of hBOA is terminated when the global optimum has been found (success) or when the upper bound on the number of generations has been reached without discovering the global optimum (failure). Note that since we use the minimum population size found by bisection, the presented results correspond only to successful runs.

To score model structures, we use the Bayesian–Dirichlet metric with likelihood equivalence [52], [53]. To bias the learning towards simpler models, we adjust the prior probability of each network to penalize complex models. We start by first computing the description length of the parameters required by the network. One parameter in the data set of size N can be accurately encoded using $0.5 \log_2 N$ bits [38]. To encode all parameters, $0.5(\sum_i |L_i|) \log_2 N$ bits are needed, where $|L_i|$ is the number of leaves in the decision tree representing parameters of variable X_i . To favor the simpler models,

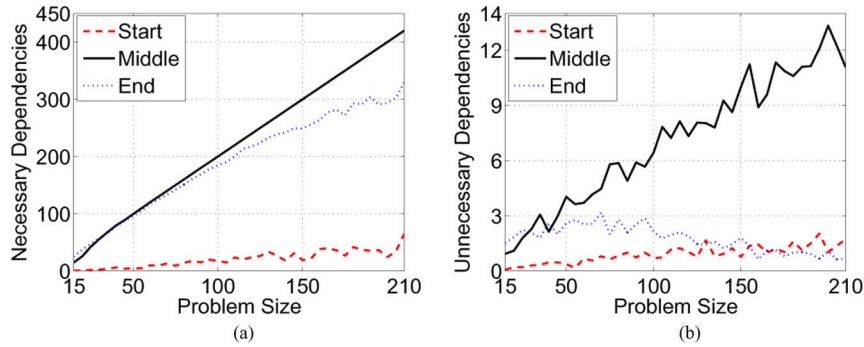


Fig. 2. The average number of necessary and unnecessary dependencies with respect to problem size on trap-5. Three model snapshots were taken in each run (the first, middle, and last generation) (a) Necessary dependencies. (b) Unnecessary dependencies.

we decrease the prior probability of each network exponentially fast with respect to the description length as suggested in [38]

$$p(B) = c2^{-0.5(\sum_i |L_i|)\log_2 N}$$

where c is a normalization constant required so that the prior probabilities of all possible networks sum to 1. The number of parents of each variable is not restricted.

D. Model Accuracy for Trap-5

We first look at the accuracy of the probabilistic models in hBOA on trap-5; specifically, we are interested in finding out how many of the necessary and unnecessary dependencies are covered. Since numerous empirical results confirmed that hBOA can solve trap-5 in a number of evaluations that grows polynomially with problem size [2], [54], we expect that the necessary dependencies will be discovered; otherwise, the deceptiveness of the trap partitions should lead to poor, exponential scalability [2], [16], [47]. On the other hand, good scalability does not directly necessitate that the number of unnecessary dependencies is relatively small compared to the overall number of dependencies. Quite the contrary, Lima *et al.* [6] showed that with tournament selection, hBOA often finds many unnecessary dependencies when solving trap-5. Nevertheless, the results with truncation selection show that the models closely resemble what we consider the perfect model for trap-5, where trap partitions are fully connected, while there are almost no connections between the different traps.

Fig. 2 shows the number of necessary and unnecessary dependencies for trap-5 of sizes 15 to 210. For each problem size, the results show the average number of necessary and unnecessary dependencies over the 30 runs using the population size determined by the bisection method. Since the models change over time, we consider three snapshots—the first snapshot shows the model in the first generation, the second snapshot shows the model in the middle of the run, and the last snapshot shows the final model learned before terminating the run (each run is terminated when the global optimum has been found).

The results in Fig. 2(a) show that initially, the probabilistic model covers only a few of the necessary dependencies. Nonetheless, the model improves over time and the second snapshot shows that in the middle of the run, all or nearly all necessary dependencies are covered (the number of necessary dependencies for trap-5 is $2n$). Finally, late in the run, the model covers many but not all necessary dependencies.

The results from Fig. 2(a) are quite intuitive. At the beginning, only some subproblems can be expected to be discovered because the initial population is generated at random and the statistical information from one round of selection does not provide enough information to identify all subproblems. However, after several more iterations, the collateral noise [16] in the population decreases, the overall quality of solutions in the population increases, and the statistical information provides enough input to enable hBOA to learn better models. Finally, at the end of the run, even though accurate problem decomposition should still be important, the dependencies between the subproblems become easy enough to cover with short-order conditional probabilities.

We illustrate the argument that in the end of the run our models can significantly simplify without affecting the encoded distribution with the following example. Consider a population of 5-bit binary strings with only two alternative candidate solutions: 00000 and 11111. Late in the run of hBOA on trap-5, every trap partition is expected to have the same or at least a very similar distribution to this. Clearly, the value of any bit depends on the value of the remaining bits. Nonetheless, to fully encode this probability distribution, we can use a simple chain model defined as

$$p(X_1, X_2, X_3, X_4, X_5) = p(X_1)p(X_2|X_1)p(X_3|X_2) \\ \times p(X_4|X_3)p(X_5|X_4).$$

Therefore, as the population diversity decreases and some partial solutions are eliminated, many of the necessary dependencies become unnecessary because they can be implicitly covered using other dependencies.

In addition to showing good coverage of necessary dependencies, the results from Fig. 2(b) show that the number of unnecessary dependencies is significantly smaller than the number of necessary dependencies. This fact is also exemplified in Fig. 3, which shows that the ratio of the number of unnecessary dependencies to the number of necessary dependencies decreases with problem size. While this may seem counterintuitive at first, as the problems increase in size and thereby contain more subproblems, larger populations are required and so while hBOA is given more dependencies to consider, it is also getting more samples of each individual subproblem. This is great news—not only does hBOA discover the dependencies we need to solve trap-5 scalably, but it is also capable of avoiding the discovery of a significant number of unnecessary dependencies, which may

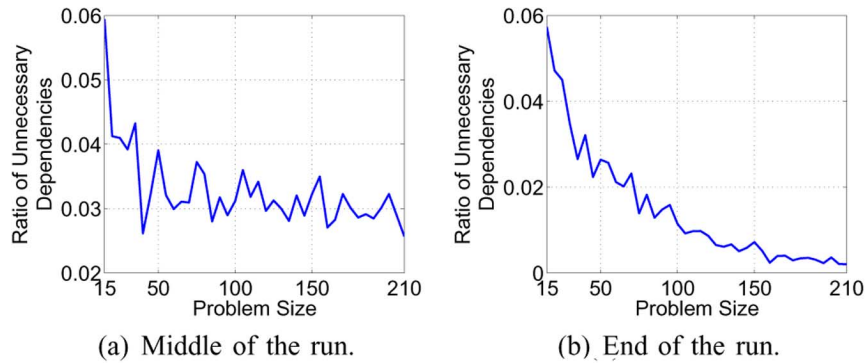


Fig. 3. Ratio of the number of unnecessary dependencies to the total number of dependencies over the lifetime for trap-5 of sizes 50–210.

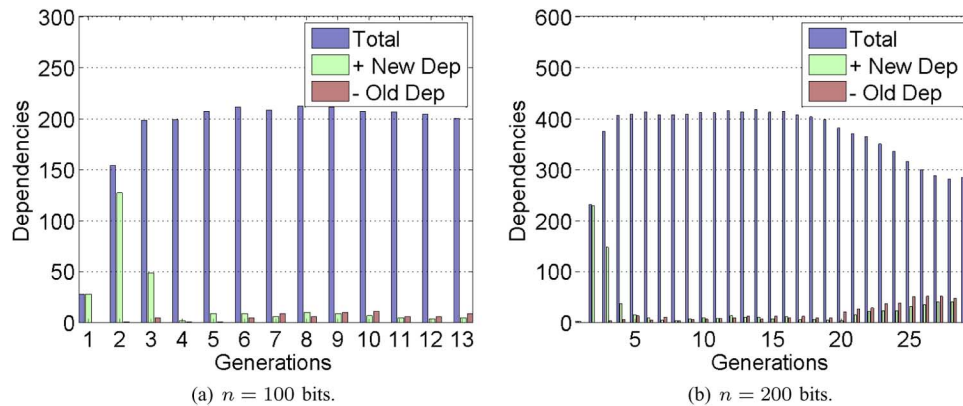


Fig. 4. Model-structure dynamics in trap-5 for two different problem sizes. For each generation, we show three bars. The first bar denotes the total number of dependencies (edges) in the current model, the second bar denotes the number of dependencies that were added in this generation but were not present in the previous one, and the third bar denotes the number of dependencies that were deleted from the model in this generation but were present in the previous generation.

slow down the mixing. However, recall that this is not always the case; as was shown elsewhere [6], the number of unnecessary dependencies becomes relatively significant with tournament selection.

One of the most surprising results obtained in this study is that the model structure in hBOA significantly depends on the selection method used to select the populations of promising solutions. More specifically, the work on hybridization of hBOA [6] as well as the work on accuracy of hBOA models with tournament selection [10] showed that with tournament selection the number of unnecessary dependencies is relatively significant and that the ratio of the number of unnecessary dependencies to the total number of dependencies increases with problem size. Furthermore, it was shown that with tournament selection nearly all necessary dependencies are discovered at the beginning of the run. Therefore, based on Lima *et al.* [6], [10], one may conclude that with tournament selection, hBOA models tend to be overly complex. On the contrary, the results presented in Figs. 2 and 3 show that for truncation selection, the models contain only a handful of unnecessary dependencies, but it takes several generations to discover all the necessary dependencies (see also Fig. 4 for additional results). Although hBOA scalability is asymptotically the same in both cases [2], [54], studying the influence of the selection method on performance of hBOA and other EDAs is an important topic and many results relevant to this topic can be found in [10].

E. Model Dynamics for Trap-5

Knowing that hBOA models for traps are accurate is great, but that is only one piece of the puzzle and there still remain many important questions. How do the models change over time? Are the models in consequent generations similar? When do most changes in models occur—is it early in the run or late in the run? All these questions are important, especially for the design of efficiency enhancement techniques, such as sporadic and incremental model building [7] and hybridization [6], [22], [55].

To better understand model dynamics in hBOA, we analyze changes in hBOA models in consequent generations of hBOA. In each generation, we record the number of dependencies that were not present in the previous generation but have been added in the current one, and analogically we record the number of dependencies that were present in the previous generation but that have been eliminated in the current one. Fig. 4 shows the obtained results for trap-5 of size 100 and 200 (the results for other problem sizes look similar). The results clearly indicate that most dependencies are added in the first few generations of hBOA. After the correct or approximately correct model for the underlying problem is discovered, the models do not change significantly.

The stability of hBOA models on trap-5 is great news for sporadic and incremental model building [7], which are efficiency enhancement techniques that focus on improving efficiency of

model building in hBOA. More specifically, both sporadic and incremental model building lead to the highest gains in performance when models in consequent generations have similar structure. Our results indicate that this is indeed the case, at least for separable problems of bounded difficulty.

While stability of hBOA models is important for incremental and sporadic model building, the rapid learning of an accurate model is also an important precondition for efficient hBOA hybrids using specialized local operators that are based on the learned models [6], [55].

IV. RANDOM ADDITIVELY DECOMPOSABLE PROBLEMS

We have shown that on separable problems with identical subproblems hBOA learns an adequate problem decomposition quickly and accurately, but what happens when we look at problems where the difficulty of each subproblem is different or the subproblems interact? Random additively decomposable problems (rADPs) [14], [15] are a particularly good type of artificial test problem for this, as they allow us to test the performance of hBOA on a broad class of nearly decomposable problems while allowing us to vary the overlap, problem difficulty and the size of the subproblems. Work has been done to show that hBOA solves rADPs scalably and robustly [14]; in this study, we want to explore how accurately hBOA divides up the problem and how the models change over time when solving rADPs.

Like in separable problems, the string positions in rADPs are partitioned into subsets of bits, with the overall fitness being the sum of the subfunctions applied to these subsets of bits. Unlike in trap-5 and other similar separable problems though, in rADPs we allow the partitions of corresponding to the different subproblems to overlap and we allow each subset of bits to contribute to the overall fitness using a different function.

Let us denote the order of rADPs by k , that is, the subproblems contain k bits each. In the simplest case, the subproblems in rADPs of order k are located in contiguous blocks of k bits and the amount of overlap is specified by a parameter o , which denotes the number of bits that are shared by neighbor subproblems. For example, for $k = 5$ and $o = 2$, the first subproblem would be located in positions (0,1,2,3,4) and the second subproblem would be located in positions (3,4,5,6,7) (positions 3 and 4 would overlap). To ensure that the subproblems are not located in contiguous blocks of bits, the positions of each rADP are shuffled using a randomly generated permutation (the shuffling remains constant during the run).

To generate a broad range of different problem instances of rADPs, for any given string length n , order k and overlap o , the fitness of each partition is given by a subfunction that is generated randomly. More specifically, the fitness for each subproblem of order k is given by a table of 2^k values (one for each possible instantiation of the corresponding k bits), which are initially generated at random. In this paper, we consider the uniform distribution from $[0, 1]$ to generate the values in the table for each subproblem as proposed in [14] and [15], but other distributions are also possible. To verify the global optimum, the algorithm described by Pelikan *et al.* [56] can be used.

Instances of rADPs are expected to vary in difficulty due to the randomness of the subfunctions. Furthermore, we can expect that the higher the value of the overlap o , the more

difficult rADPs may get, because the interactions between the different subproblems can become stronger if the subproblems share more bits.

In this paper, we consider rADPs of order $k = 5$ and with overlap $o = 0, 1$, and 2 . All instances are generated using the implementation described in [57].

A. Perfect Model for rADPs

Unlike in trap-5, the perfect model for rADPs depends on the subfunctions corresponding to the different subproblems. Without any assumptions about the different subproblems, a perfect model should encode dependencies between all pairs of bits in each subproblem, leading to a model similar to that used for solving trap-5. In fact, any subproblem can be deceptive, which would indeed require hBOA to use a model that covers dependencies between all pairs of bits in this subproblem. This can also be verified using the theory presented in [46]. However, for some subproblems, the contributions of different bits may be nearly independent, which would mean that the model corresponding to this subproblem does not have to contain any dependencies.

Therefore, for the most difficult subproblems with strong interactions between the bits, the model should contain dependencies between all or nearly all bits in the subproblem, whereas for the simplest subproblems the model does not have to encode nearly any dependencies at all.

Just as in the previous section, we will call dependencies between bits in each subproblem *necessary* and the remaining dependencies *unnecessary*. It must be understood though that some of the necessary dependencies may not be required for the easier subproblems; thus, the distinction between the necessary and unnecessary dependencies is not as sharp as in trap-5.

B. Experimental Setup

As mentioned in the previous section, we considered rADPs with $k = 5$ and $o = 0, 1$, and 2 . In order to examine how the models change as problem size increases, for each overlap we examined problems containing from 5 to 25 subproblems (in the case of $o = 2$ we also considered larger problems). Since rADP vary in difficulty, the experiments were performed on 1000 different instances for each problem size and overlap, with all problem instances generated by the code provided by Pelikan *et al.* [57]. With three levels of overlap and five problem sizes for all but the largest overlap (where we looked at one additional size), we examined 16 000 instances.

Most hBOA parameters were set the same as in the concatenated traps. The population size for any given instance was determined by bisection, with each run required to converge in ten out of ten independent runs.

C. Model Accuracy for rADPs

We start our examination of model accuracy in hBOA on rADPs by first looking at problems of overlap 0. To reduce the number of results, we only look at the models in the middle of the run. Fig. 5(a) and (b) show the number of necessary and unnecessary dependencies for rADPs of sizes 25 to 125 with no overlap. Since we wanted to examine how model accuracy

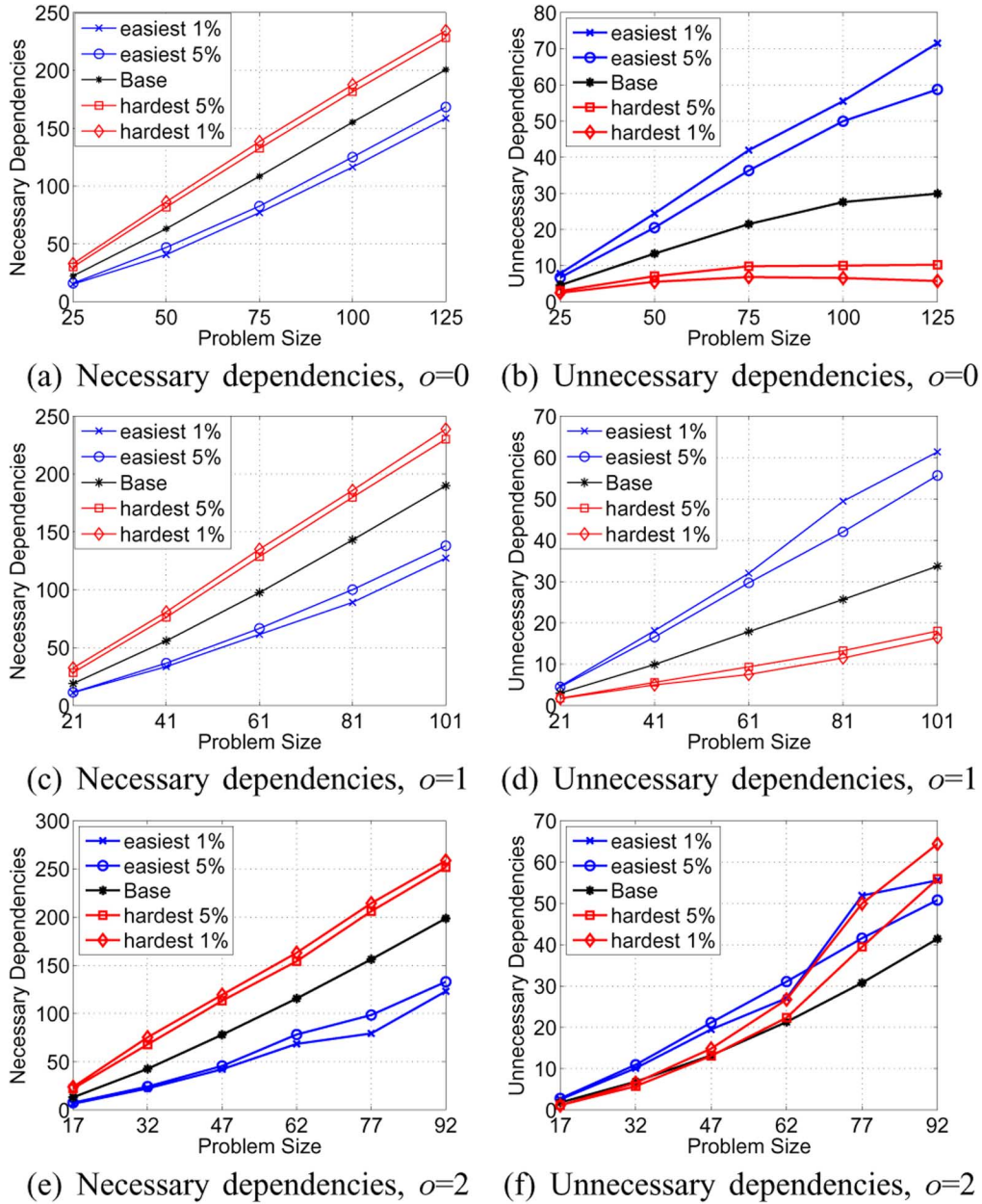


Fig. 5. Average number of necessary and unnecessary dependencies with respect to problem size for rADP with various overlaps at the middle of a run. Five different levels of difficulty are considered.

changes as problem difficulty varies, we show average results for all 1000 problem instances for each problem size, and then the easiest and hardest 5% and 1% of the instances (with difficulty determined by the number of evaluations required by hBOA to solve the problem).

As argued earlier, we expected that for the hardest problems, most of the necessary dependencies would be required and the necessary population size would also be expected to be relatively high compared to the easiest problems. The results from Fig. 5(a) confirm this hypothesis. While it might at first seem that we are missing some necessary dependencies even in the hardest problems, what must be remembered is that even in the hardest problems, some of the subproblems may still be easier and the subproblems may not be equally scaled, leading to sequential (domino-like) convergence [58]. That is why it can be

expected that even in the hardest problems the number of dependencies is slightly smaller than that in trap-5.

If the population size would be increased also in the easiest instances, the probabilistic models would be expected to cover more of the dependencies in the underlying problem. The reason for this is that the number of dependencies covered by the probabilistic model can be expected to be correlated with the population size [44], [51]. This point is discussed later in this section (see also Fig. 7).

Fig. 5(b) shows that as expected, the number of unnecessary dependencies discovered is significantly smaller than the number of necessary dependencies found. Furthermore, for harder problems, the ratio of the number of unnecessary dependencies and the number of necessary dependencies decreases with problem size. So, not only do we discover the necessary

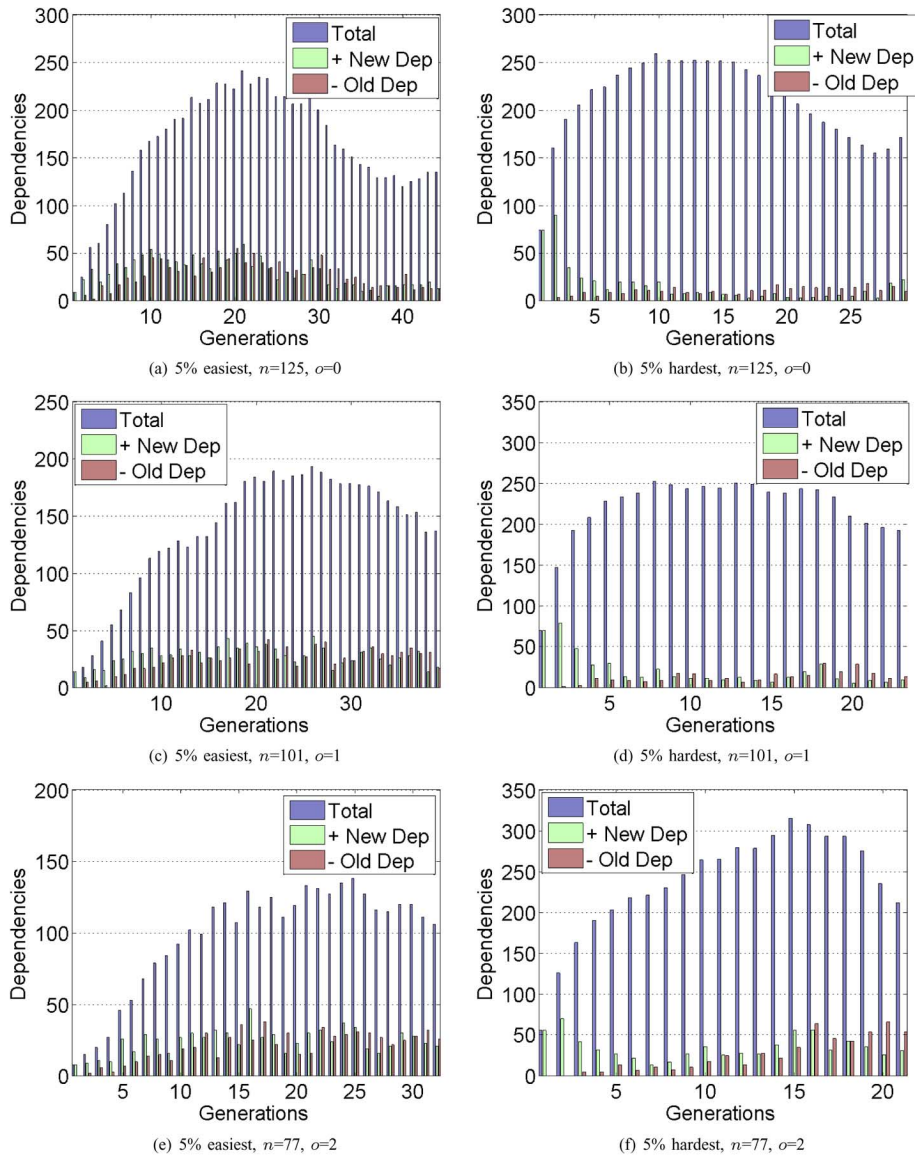


Fig. 6. Model-structure dynamics for selected runs of rADPs of varying difficulty. Similarly as in the results for trap-5, for each generation, we show three bars. The first bar denotes the total number of dependencies (edges) in the current model, the second bar denotes the number of dependencies that were added in this generation but were not present in the previous one, and the third bar denotes the number of dependencies that were deleted from the model in this generation but were present in the previous generation.

dependencies, but again as in trap-5, hBOA avoids finding significant numbers of unnecessary dependencies, at least in the harder problems. While hBOA does find more unnecessary dependencies in the easiest problems, this is to be expected as in those problems nearly all dependencies are in fact unnecessary and the populations are relatively small.

Next, we look at problems of overlap 1. Fig. 5(c) and (d) show the number of necessary and unnecessary dependencies for problem sizes 21 through 101. We see much the same pattern as in problems with overlap 0, at least with regard to the necessary dependencies. On the other hand, we see a much more stable ratio of unnecessary to necessary dependencies than for overlap 0, without the decrease in the ratio that we saw with the completely independent subproblems. The reason for this behavior is that the overlap introduces interactions between bits in the different subproblems and the strongest interactions get re-

flected by the model despite that they are unnecessary. However, this ratio is still small and hBOA is not finding large numbers of unnecessary dependencies even with overlap 1.

Finally, we look at problems of overlap 2. Fig. 5(e) and (f) show the number of necessary and unnecessary dependencies for problem sizes 17 through 92. As in the previous cases, by the middle of the run, even for heavy overlap, hBOA has found most of the necessary dependencies. While for all sizes shown the number of unnecessary dependencies shown is small, the ratio of unnecessary to necessary dependencies seems to increase as problem size gets bigger. This result is not surprising—the stronger the overlap, the more likely the population exhibits statistical dependencies between the bits in the different subproblems. This is mainly because, unlike in separable problems, in rADPs with overlap the subproblems are not independent. While the dependencies between different subproblems are ex-

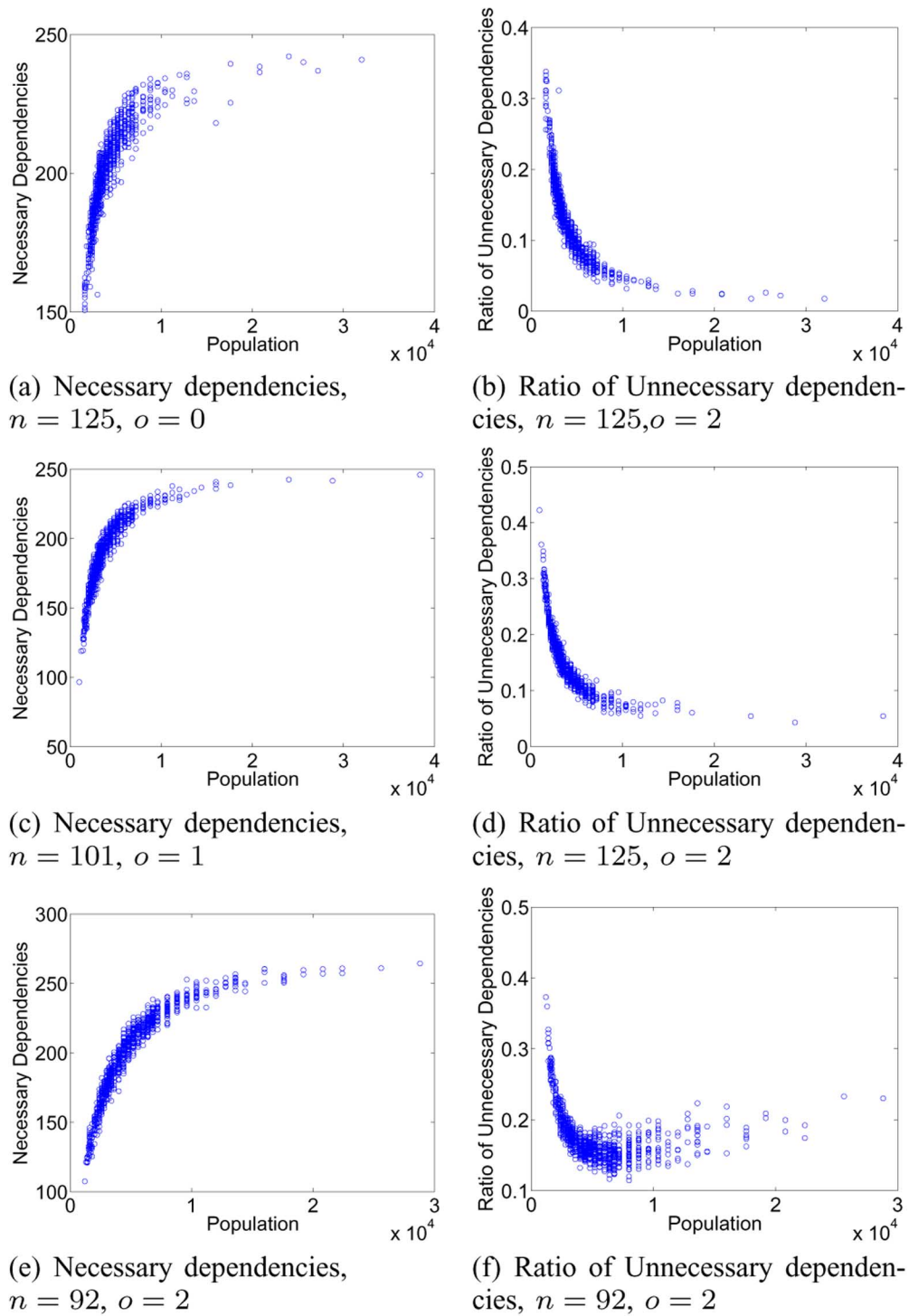


Fig. 7. Average number of necessary dependencies and the ratio of the number of unnecessary dependencies to the overall number of dependencies at the middle of the run with respect to the population size. One thousand random problem instances of one size for each overlap are considered.

pected to be weaker than those within a subproblem, they still may be strong enough to get covered by the probabilistic model in some cases.

Yet another question arises. How do the models found by hBOA change as the population size changes? To examine this question, in Fig. 7, we look at the number of necessary dependencies found in the middle of the run with respect to population size for all instances of largest problem sizes. In addition, we

also look at the ratio of unnecessary dependencies to total dependencies with respect to the population size during the middle of the run for all instances. All these figures show similar results. As the population sizes get larger, more necessary dependencies are discovered. We also see that for the easier instances (those where bisection found smaller population sizes), much fewer necessary dependencies were found, as in these simpler problems, fewer dependencies were required.

D. Model Dynamics for rADPs

While we have looked at the structure of the models for rADPs in the middle of the run, it is also important to examine how the models change over time, especially when we consider that for rADPs the subproblems vary in difficulty. To examine this, we look at the number of dependencies that change and the number of dependencies that remain the same from generation to generation. Fig. 6 shows six selected runs, with each one selected from a different difficulty and overlap. Although we are only showing one run for each problem type, other runs of the same type were similar and the presented results are thus indicative of the general pattern of model dynamics of that type.

For the harder instances, just like in trap-5, we see that hBOA adds most of the dependencies early on in the run. After the first few generations, the models are relatively stable, changing only a little at a time. Near the end of the run, as individual subproblems are solved, they can be modeled with short-order probabilities and so the overall number of dependencies decreases. For the easier problems, we see a ramping up of dependencies as the earlier generations progress, but the models are still relatively stable by the middle of the run. All this is good news, as even in the case of hard problems with heavy overlap, hBOA is able to find a relatively stable model quickly and the models remain relatively stable (their structure does not change fast).

V. HBOA MODELS ON HIERARCHICAL PROBLEMS

The problems that we have looked at so far, while all difficult in different ways, all shared a common characteristic: they can be solved by a fixed decomposition at a single level. While many difficult problems can be decomposed in some way, using a single-level decomposition may not be enough; instead it might be necessary to use hierarchical decomposition [2], [16], [43] and compose solutions over multiple levels of difficulty. hBOA is capable of solving difficult hierarchical problems in a robust and scalable manner [2], [16], [44]. When solving hierarchical problems, hBOA starts at the bottom level with subproblems of small order and at each level of optimization, hBOA uses promising solutions of the subproblems at the lower level as basic building blocks to juxtapose solutions at the current level.

While it has been empirically shown that hBOA can solve difficult hierarchical problems [2], [16], [44], the same questions as for single-level decomposable problems arise: What do the models in hBOA look like when solving hierarchical problems and how do these models change over time? In this section, we try to answer these questions using hierarchical deceptive traps as the basic problem. Hierarchical traps were chosen because they bound the class of hierarchical problems, they are deceptive at all levels, and they are highly multimodal [2], [16].

A. Hierarchical Traps of Order 3

Hierarchical traps (hTraps) [1], [2] of order 3 are created by combining trap functions of order 3 over multiple levels of difficulty. In the variant of hTrap used in this work, on the lowest level, groups of 3 bits contribute to the overall fitness using a generalized 3-bit trap

$$\text{trap}_3(u) = \begin{cases} f_{\text{high}}, & \text{if } u = 3 \\ f_{\text{low}} - u \left(\frac{f_{\text{low}}}{2} \right), & \text{otherwise} \end{cases} \quad (3)$$

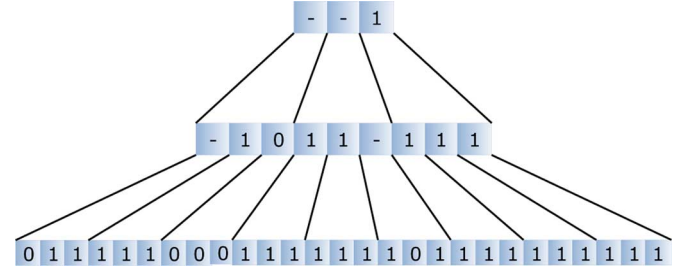


Fig. 8. An illustration of the mapping function for hTrap of order 3 with three levels. A 000 is mapped to a 0, a 111 is mapped to a 1, and everything else is mapped to the null symbol “—”.

where $f_{\text{high}} = 1$ and $f_{\text{low}} = 1 + 0.1/l$. Note that for these values of f_{high} and f_{low} the optimum of the trap is 000.

Each group of 3 bits corresponding to one of the traps is then mapped to a single symbol on the next level; a 000 is mapped to a 0, a 111 is mapped to a 1, and everything else is mapped to the null symbol “—”. See Fig. 8 for an example of this mapping function. The bits on the next level again contribute to the overall fitness using 3-bit traps defined above, and the groups are mapped to an even higher level. This continues until the top level is evaluated that contains 3 bits total. However, on the top level, a trap with $f_{\text{high}} = 1$ and $f_{\text{low}} = 0.9$ is applied. As a result, the optimum of hTrap is in the string of all ones despite the superior performance of blocks of zeros on any level except for the top one. Note that any group of bits containing the null symbol does not contribute to the overall fitness.

To ensure that the total contribution from each level is of the same magnitude, the contributions of traps on the l th level from the bottom are multiplied by 3^l .

While hTrap has many local optima, the global optimum is the string all 1s. Nonetheless, any single-level decomposition into subproblems of bounded order will lead away from this global optimum. So hTrap requires an optimizer that can build hierarchically on good partial solutions over multiple levels of difficulty until the global optimum is found.

B. Perfect Model for Hierarchical Traps of Order 3

Both trap-5 and rADPs can be solved using a fixed decomposition of bounded order and we have shown that such a decomposition is found quickly and reliably. However, to solve hTraps, a fixed decomposition is insufficient because any decomposition of bounded order will be misled away from the global optimum and using a decomposition of order growing with problem size will result in exponential complexity. To solve hTrap scalably, hBOA should proceed level by level, first finding optima of subproblems on each level and then proceeding to the next level where these optima would be juxtaposed to form partial solutions of higher order.

More specifically, on the lowest level, hBOA should first find the local optima of the 3-bit traps on this level, setting the subset of 3 bits corresponding to each trap either to 000 or 111. As most of the subproblems on the lowest level would converge to either 000 or 111, the subproblems on the second level could be solved leading to the blocks of nine 0s or nine 1s. In this manner, hBOA should compose solutions of higher and higher order, always encoding blocks of 0s and 1s.

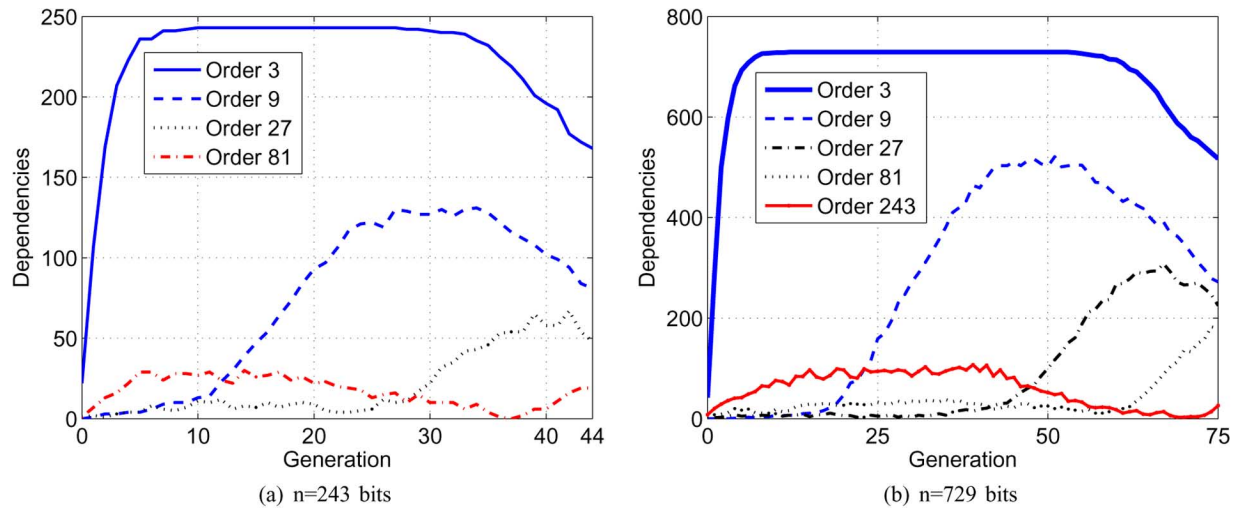


Fig. 9. The number of dependencies of different orders in one hBOA run on hTrap of sizes $n = 243$ and $n = 729$. Other runs led to similar results.

To analyze the models of hBOA for hTraps, we thus look at the dependencies between bits in each subproblem on different levels of the hierarchy. Dependencies between the bits on the lowest level will be referred to as dependencies of order 3 as they connect bits within 3-bit subproblems. Dependencies on the second level from the bottom will be referred to as dependencies of order 9 as they connect bits within 9-bit subproblems; order-3 dependencies will be excluded from order-9 dependencies. Analogically, the dependencies of order 3^l for any level l will denote the dependencies between the 3^l bits of the subproblems on the l th level from the bottom, excluding dependencies of order $3^1, 3^2, \dots, 3^{l-1}$.

As hBOA solves an hTrap, it should start by discovering dependencies of order 3. Next, it should discover dependencies of order 9. In this manner, it should discover dependencies of higher and higher order until the global optimum is found and the run is terminated.

C. Experimental Setup

Most parameters were set the same as for concatenated traps. Population size was determined by bisection, with hBOA required to converge in 30 out of 30 independent runs. In order to look at the probabilistic models of hBOA on hTrap as the problem size increased, we looked at problems of sizes $n = 243$ and $n = 729$.

D. Model Dynamics for Hierarchical Traps

Fig. 9 shows the number of dependencies of different orders at various stages of a sample run; the results in other runs were similar and were thus omitted. Early on in the run, hBOA quickly adds dependencies, encoding almost all of the dependencies of order 3, which should be sufficient to solve the problem at the first level. Yet to start with, very few dependencies of higher order are included.

As hBOA continues to the next level, it has to encode blocks of nine 0s or 1s in order to encode partial solutions on the second level. Indeed, we see that at around generation 10 for hTrap of $n = 243$ bits and at around generation 15 for hTrap of $n = 729$ bits there is a rapid increase in the number of dependencies

of order 9, indicating that the optimization has proceeded to the second level. Next, hBOA proceeds to the third level as is supported by the rapid increase in the number of dependencies of order 27, starting at around generation 27 for $n = 243$ and at around generation 45 for $n = 729$. The same pattern can also be seen for $n = 729$ and dependencies of order 81, which correspond to level 4.

It is important to note that as we proceed to higher levels, only a few dependencies need to be added as the subproblems that converge to only a few local optima can be encoded with a small number of dependencies, similarly as in trap-5 and rADPs. Specifically, if we want to encode all dependencies between all bits in one block of either 0s or 1s and those in another block of 0s or 1s, it is sufficient to cover all such dependencies by connecting any bit from the first block with any bit of the second block; all other dependencies between the two blocks will be covered implicitly if the two blocks are properly covered by the model. Note also that the increase in the number of dependencies at one before the last level is not as clear because only a few of these dependencies are necessary to finalize solving the problem and some of these dependencies are already included in the model throughout the run due to the imperfect information from the finite population size.

The most striking thing we notice from these results is the regularity of the hierarchical search process. We argued earlier that hTraps cannot be solved using a fixed decomposition throughout the entire search. Indeed, the results show that the model changes at sharply defined stages—after hBOA finishes solving the subproblems at one level, it moves on to decomposing the next level. Also of note is that hBOA does keep the small order dependencies throughout almost all of the run, which is necessary as we must preserve alternative solutions (and their structure) in order to solve higher levels. The regularity of this process is good news for sporadic and incremental model building, as new structures need only be discovered as we transfer from one level to the next one.

Furthermore, since the models experience rapid change when one level has been decomposed and the next level starts being partitioned, the results imply that hBOA can be used to educate

researchers about their hierarchical problems, by providing information about the hierarchical structure of the problem and the way in which this structure can be used to find the global optimum.

VI. HBOA MODELS FOR TWO-DIMENSIONAL SPIN GLASSES

We have shown on difficult separable problems of bounded order hBOA learns an adequate problem decomposition quickly and accurately, and even if the problem has to be solved hierarchically, hBOA finds a good decomposition at each level of difficulty and solves the problem level by level. However, all considered problems are artificial test problems with a relatively simple structure, and with enough information about these problems it is straightforward to find adequate probabilistic models. What happens when we move to real-world problems where the structure is not as easy to crack and where it is unclear what a good probabilistic model should look like? Will we still be able to learn something about the problem from the probabilistic models discovered by hBOA and will these models closely correspond to the underlying problem? Will the models change much during the run?

To answer this question, we consider the problem of finding ground states of two-dimensional Ising spin glasses [17]–[20], [22], [25], [59]–[61], where the task is to find spin configurations that minimize the energy of a given spin glass instance. While the structure of the energy function for two-dimensional Ising spin glasses is relatively easy to understand, the problem of finding spin-glass ground states represents a great challenge for most optimization techniques because: 1) the energy landscape contains an extremely large number of local optima; 2) the local optima in the energy landscape are often separated by configurations with very high energy (low-quality regions); 3) any decomposition of bounded order is insufficient for solving the problem; and 4) the problem is still solvable in polynomial time using analytical methods. This is why most standard and advanced optimization techniques fail to solve this problem in polynomial time, including classical genetic algorithms [22] and state-of-the-art Markov chain Monte Carlo (MCMC) methods such as the flat-histogram MCMC [24]. Despite the difficulty of the Ising spin glass problem, hBOA is able to solve two-dimensional Ising spin glasses with $\pm J$ couplings in polynomial time, achieving empirical asymptotic performance of the best analytical methods without any problem-specific knowledge [2], [22], [59].

A. Two-Dimensional Ising Spin Glass

A very simple model to describe a finite-dimensional Ising spin glass is typically arranged on a regular two-dimensional or three-dimensional grid, where each node i corresponds to a spin s_i and each edge $\langle i, j \rangle$ corresponds to a coupling between two spins s_i and s_j . For the classical Ising model, each spin s_i can be in one of two states: $s_i = +1$ or $s_i = -1$. Each edge $\langle i, j \rangle$ has a real value (coupling) $J_{i,j}$ associated with it that defines the relationship between the two connected spins. To approximate the behavior of the large-scale system, periodic boundary conditions are often used that introduce a coupling between the first and the last element along each dimension.

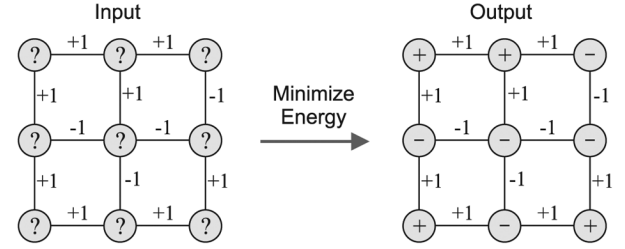


Fig. 10. Finding ground states of Ising spin glasses in two dimensions. For simplicity, we omit periodic boundary conditions in this figure.

In this paper, we consider the two-dimensional Ising spin glass with periodic boundary conditions.

Given a set of coupling constants $J_{i,j}$, and a configuration of spins $C = \{s_i\}$, the energy can be computed as

$$E(C) = \sum_{\langle i,j \rangle} s_i J_{i,j} s_j \quad (4)$$

where the sum runs over all couplings $\langle i, j \rangle$. Here, the task is to find a spin configuration given couplings $\{J_{i,j}\}$ that minimizes the energy of the spin glass; see Fig. 10 for an illustration. The states with minimum energy are called *ground states*. In hBOA, the spin configurations are encoded with binary strings, where each bit specifies the value of one spin (0 for +1, 1 for -1).

In order to obtain a quantitative understanding of the disorder in a spin glass system introduced by the random spin–spin couplings, one generally analyzes a large set of random spin glass instances for a given distribution of the spin–spin couplings. For each spin glass instance, the optimization algorithm is applied and the results are analyzed. Here, we consider the $\pm J$ spin glass, where each spin–spin coupling constant is set randomly to either +1 or -1 with equal probability. All instances of sizes up to 18×18 with ground states were obtained from S. Sabhapandit and S. N. Coppersmith from the University of Wisconsin who identified the ground states using flat-histogram Markov chain Monte Carlo simulations [24]. The ground states of the remaining instances were obtained from the Spin Glass Ground State Server at the University of Cologne [62].

To improve the performance of hBOA, we incorporated a deterministic local search—deterministic hill climber (DHC)—to improve quality of each evaluated solution [2], [22]. DHC proceeds by making single-bit flips that improve the quality of the solution most until no single-bit flip improves the candidate solution. DHC is applied to every solution in the population before it is evaluated.

B. Perfect Model for Two-Dimensional Ising Spin Glass

For two-dimensional spin glasses, it is not clear what dependencies are really necessary to solve the problem scalably and neither is it clear what dependencies are unnecessary and should be avoided to maximize the mixing. In fact, to some degree, every spin (bit) depends on every other spin either directly through one connection (coupling) or through a chain of connections. It has been argued [21] that to achieve provable convergence on two-dimensional spin glasses, the order of dependencies in the probabilistic model must grow at least as fast as $\Omega(\sqrt{n})$. Of course, since hBOA has been empirically shown

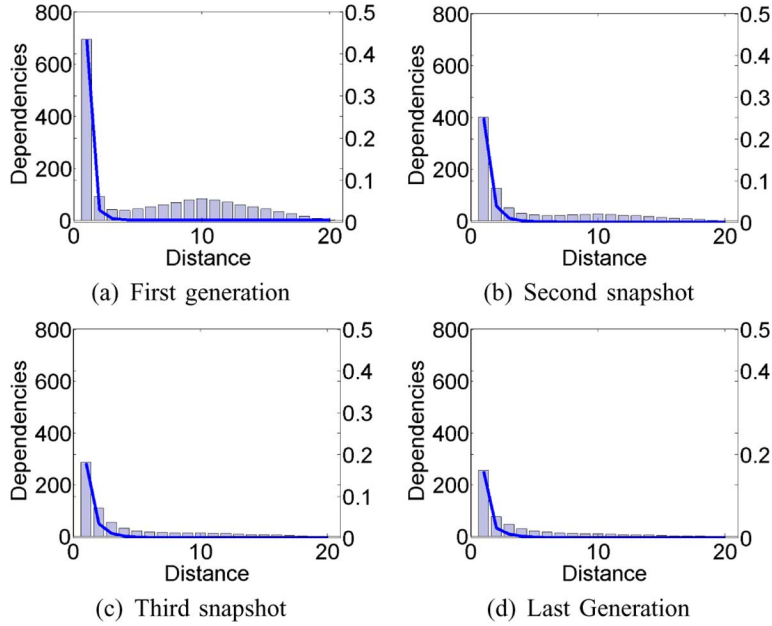


Fig. 11. Histogram of the distribution of dependencies with respect to spin distances for hBOA with DHC. The curve displays the number of dependencies normalized by the total number of dependencies considered by hBOA at any distance.

to solve two-dimensional Ising spin glasses in polynomial time and due to the initial-supply population sizing [16] the population size in hBOA is lower bounded by $\Omega(2^k)$ (where k is the order of dependencies covered by the probabilistic model), the models in hBOA are not expected to encode dependencies of order $\Omega(\sqrt{n})$ or more. For more details on the empirical results supporting polynomial scalability of hBOA on two-dimensional spin glasses, please see refs. [2], [22], [25], and [59].

Nonetheless, while it is unclear what a “perfect” model for a two-dimensional Ising spin glass looks like, it is clear that the interactions between immediate neighbors should be strongest and the interactions should decrease in magnitude with the distance of spins measured by the number of links between these spins (considering the shortest path of links between the spins). This hypothesis will be confirmed with the experimental results presented shortly.

C. Experimental Setup

Most hBOA parameters were set the same as in the experiments for concatenated traps. Since the difficulty of spin glass instances varies significantly depending on the couplings [22], [24], we generated 100 random instances for each considered problem size. On each spin glass instance, we used the population size obtained by bisection but since we used 100 random instances for each problem size, we required hBOA to converge only in five out of five independent runs.

D. Model Dynamics and Structure for Two-Dimensional Ising Spin Glass

Given our initial understanding of the structure of two-dimensional Ising spin glasses, we start by analyzing the grid distances of the spins directly connected by a dependency in the probabilistic model, where the distance of two spins is defined as the minimum number of links in the two-dimensional grid that we

must pass to get from one spin to the other one. The minimum distance, one, is between any immediate neighbors in the grid. Due to the periodic boundary conditions, the maximum distance is *not* between the opposite corners of the grid. Instead, the maximum distance is, for example, between the top-left spin and the spins around the middle of the grid; for instance, for two-dimensional spin glasses of size 20×20 the maximum distance is 20 (10 links to the right, 10 links to the bottom). Also of note is that the number of possible dependencies at any given distance increases to a maximum at half of the maximum distance. For example, any particular spin in a two-dimensional spin glass of size 20×20 could have 38 possible dependencies of distance 10 but only 1 dependency possible of length 20.

To analyze distances of spins that are connected in the probabilistic model, we created a histogram of the average number of dependencies for each distance at different stages of the hBOA run on 100 randomly generated $\pm J$ Ising spin glasses. We expected that early in the run, majority of the dependencies would be between immediate neighbors in the grid, whereas later in the run, the number of dependencies between further spins would increase. We obtained four snapshots of the model at equally distributed time intervals to cover the entire run of hBOA.

Fig. 11 shows that in all stages of the run, the number of dependencies between immediate neighbors is about as large as the total number of the remaining dependencies. Since the influence of each spin on the distribution of another spin decreases significantly with the number of links between the spins, we believe that (at least initially) the dependencies of spins that are located far away are not necessary and are only a result of the strong problem regularities and noisy information of the initial population of spin configurations. However, as the run proceeds, long-range dependencies may become more meaningful as the search progresses to locate the candidate solutions in the most promising regions of the search space.

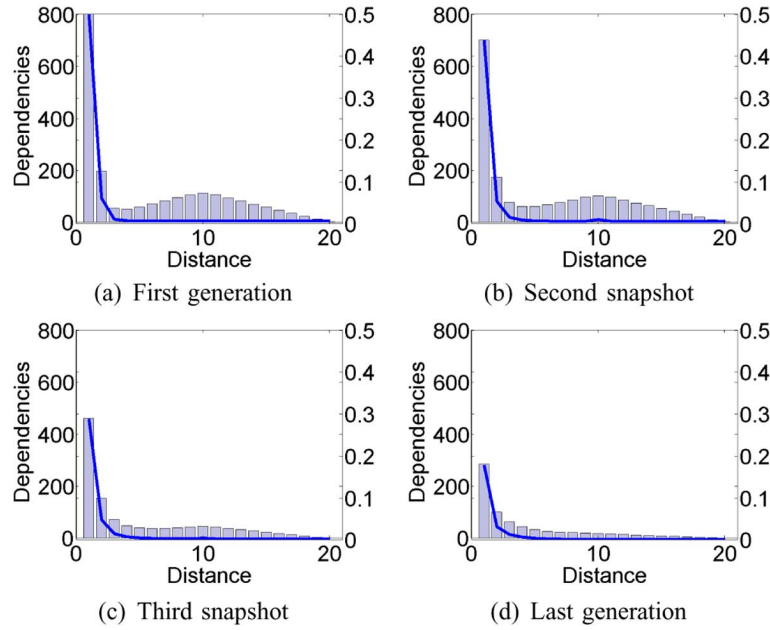


Fig. 12. Histogram of the distribution of dependencies with respect to spin distances for hBOA without DHC. The curve displays the number of dependencies normalized by the total number of dependencies considered by hBOA at any distance.

One of the surprising results from Fig. 11 is that the number of dependencies at any distance decreases over time and, therefore, in each generation of hBOA, the model becomes simpler. We believe that while the effective order of dependencies encoded by the model increases over the run, since these dependencies become simpler (they cover fewer alternative partial solutions), they can be covered with structures composed of short-order dependencies. This behavior is similar to that observed for concatenated traps in the last stage of the run, where some necessary dependencies were not really needed. Also of note is the increased number of mid-range dependencies that exist early on. We believe this is a result of the increased total number of dependencies possible at these distances, some of which are found simply due to the use of relatively small, finite populations. To illustrate this point, each of the histograms has an overlaid plot of the number of dependencies found normalized by the number of possible dependencies at that distance. We see from these plots that in terms of the proportion of dependencies covered at each distance, the increased number of mid-range dependencies disappears. Note that we are normalizing by all possible dependencies that could be considered. In practice, many of these dependencies would be ruled out during the run to avoid cycles in the resulting model. For example, any spin has four possible neighbors to consider but the maximum possible dependencies at distance one for any spin glass model would be twice the number of spins.

To check whether the spin glass results are significantly influenced by DHC, we repeated the same experiments without DHC (see Fig. 12). The results without DHC indicate that the structure of the models looks similar to that found with DHC; however, the long-range dependencies become even more significant without DHC than with DHC. In fact, without DHC, the models are generally more complex. This is most likely a result

of the increased diversity in the population, which leads to more complex models.

Just as with trap-5, we next look at how stable individual dependencies are in consequent generations. In each generation, we record the number of dependencies that were not present in the previous generation but have been added in the current one and, analogically, we record the number of dependencies that were present in the previous generation but that have been eliminated in the current one. Fig. 13(a) shows the obtained results for one run of a 20×20 spin glass (the results for other runs are similar). The results seemed to indicate that the models change rapidly. However, in Fig. 13(b), we repeated the experiment and only looked at neighbor dependencies. These results clearly show that the short-range dependencies are very stable and the changes seen in Fig. 13(a) are due to the farther dependencies changing. Since many long-range dependencies can be expected to be unnecessary as is also supported by the results presented in Section VI-E, the most important factor regarding the stability of hBOA models is the stability with respect to the short-range dependencies. An additional reason for the importance of the stability of short-range dependencies is that long-range dependencies can be expressed in many different ways, especially once blocks of strongly correlated spins are located next to each other. Nonetheless, like for hTraps, for spin glasses it appears to be important for the models to change over time since two-dimensional spin glasses are complex problems that might require hierarchical problem solving.

E. Restricting hBOA Models on Spin Glass

We have observed that most of the dependencies hBOA finds are the short dependencies between neighbors in the grid. So the following question arises, if restricted to only dependencies of distance one, would hBOA still solve the problem scalably and

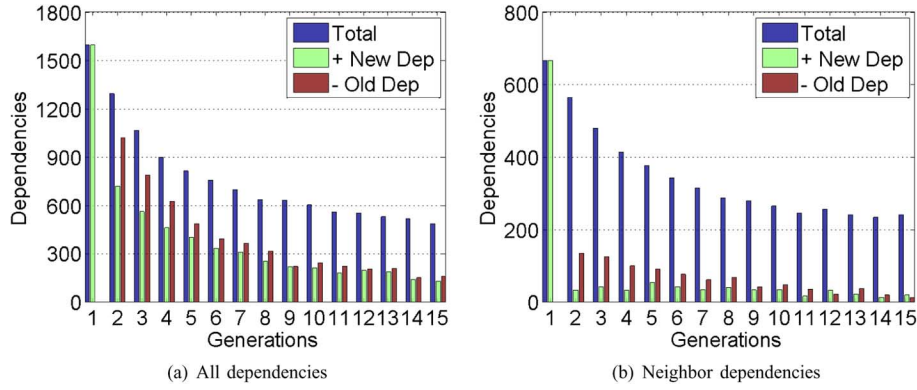


Fig. 13. Model-structure dynamics for one run on a 20×20 spin glass. Similarly, as in the results presented earlier, for each generation, we show three bars. The first bar denotes the total number of dependencies (edges) in the current model, the second bar denotes the number of dependencies that were added in this generation but were not present in the previous one, and the third bar denotes the number of dependencies that were deleted from the model in this generation but were present in the previous generation.

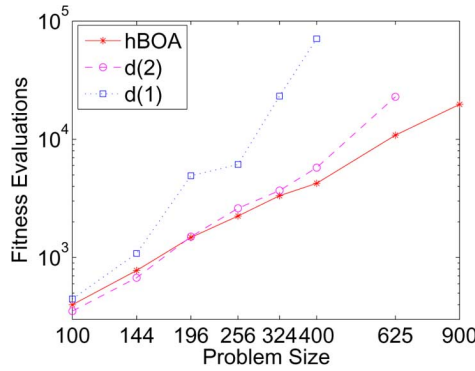


Fig. 14. Scalability of hBOA with DHC with and without restrictions on model structure.

reliably? If not, could we at least restrict the dependencies to consider distances of at most two or three? If we could restrict hBOA models significantly without negatively affecting hBOA scalability, these restrictions could be used both to significantly speed up hBOA as well as to design powerful problem-specific recombination operators for spin glasses and other similar problems.

To answer the above question, we looked at the scalability of hBOA with the different restrictions on the dependencies using spin glass instances of sizes from $10 \times 10 = 100$ spins to $30 \times 30 = 900$ spins, using 100 instances for each problem size. For each problem instance and each hBOA variant, we used bisection to find an adequate population size and recorded the average number of evaluations until convergence. The results are shown in Fig. 14.

The results show that while performance of hBOA restricted to neighbor dependencies is comparable to the performance of the original hBOA on the smallest spin glass instances of size 10×10 , as the problem size increases, performance of hBOA with only neighbor dependencies deteriorates quickly and the number of evaluations starts to grow faster than polynomially.

As also shown in Fig. 14, performance of hBOA with dependencies restricted to spins at distances of at most two leads to performance comparable to that of the original hBOA up to the problems of size 18×18 . However, as the problem size is fur-

ther increased, the performance of the restricted hBOA again starts to deteriorate, and some instances of size 30×30 were not solved even with a population size of 500 000.

The scalability analysis thus indicates that to solve two-dimensional spin glasses scalably we cannot restrict the dependencies to be only between the immediate neighbors or spins located at the distance of at most two. This result agrees with the results presented in [63] and it also shows that the hand-designed approximate Bayesian network structure proposed for two-dimensional Ising spin glasses in [21] is unlikely to scale up because it relates only the spins at distance one or two. Nonetheless, the results show that we can restrict hBOA models significantly although the degree to which we can restrict the models appears to depend on the problem size, assuming that we do not want to affect hBOA scalability. Because of the important implications of restricting complexity of hBOA probabilistic models, this topic should be further investigated in future research.

The implications of these results go beyond the direct application of hBOA to spin glasses. Specifically, we show that if one restricts the models in two-dimensional spin glasses based on prior information about the problem to only contain interactions between directly connected spins or even pairs of spins at distance two, performance of hBOA becomes intractable for large problems. Thus, while the probabilistic model based on the running intersection property leads to intractable performance due to the large order of dependencies it uses [21], the approximate probabilistic models restricted according to the structure of the underlying problem [21] fail to scale up for large problems. Thus, in spite of the availability of the complete knowledge of the problem specifics, it is not straightforward to restrict the probabilistic models appropriately or design scalable probabilistic models by hand.

It is important to note, however, that for some problems, simple probabilistic models are sufficient even though the structure of the objective function is relatively complex (suggesting the need for complex multivariate models) [64]–[66].

VII. RELATED WORK

One of the first attempts to look at the internal working of Bayesian networks in EDAs was done in [67]. In this study, the

LFDA [46] was compared against several other algorithms on the graph bipartitioning problem. Like hBOA, LFDA also builds and samples Bayesian networks, although LFDA uses the BIC metric [68] to score network structures and unlike hBOA it does not use local structures or niching. To speed up model building and provide more accurate models, LFDA was restricted to only add edges between variables that corresponded to actual edges in the problem graph. The study showed that in the first generation LFDA added much fewer edges than the maximum number of allowed edges, and as the population converged, the number of edges in the models dropped. However, no results were shown without model restrictions and the study was limited to only a few instances of one problem class.

The models built by LFDA were again examined in [69]. This time the models generated by LFDA when solving three benchmark problems were looked at. In two of the problems, the models closely matched the underlying structure. However, in the case of a deceptive problem of order 5, LFDA had trouble finding good dependencies except when given a large population size. The authors theorized that this was due to the difficulty of representing all the dependencies between variables given the constraint on the maximum number of incoming edges (which was set to eight in these experiments). While these results are somewhat in contrast with ours—here we show that hBOA finds accurate models on concatenated traps of order 5—in hBOA we use a different metric to score network structures, we use local structures to represent Bayesian networks more efficiently, and we do not enforce any constraint on the maximum number of incoming edges.

Bengoetxea [70] in his thesis looked at the evolution of graphical models in solving the graph-matching problem. He showed that the EBNA [71], which also uses Bayesian networks, started with a fairly complex network that would decrease in complexity as the population converged, ending with a model with no dependencies. He did not discuss how many of these edges were necessary and how many were unnecessary for solving the problem. One interesting result presented in this study was that in the continuous domain using Gaussian networks in the estimation of Gaussian networks algorithm (EGNA) [28], [72], the trend was actually reversed, with the model gaining complexity throughout the run. However, the study considered only a very small number of instances.

Finally, Santana *et al.* [73] used three different EDA models to solve the protein folding problem. They then examined how accurately these models corresponded to the underlying problem structure. Their results showed that the models did indeed mimic some features of the underlying problem. In addition, they also discussed how incorporating prior knowledge when solving the protein folding problem could be useful in an EDA.

If we know the structure of the objective function, it is possible to design a probabilistic model that will provably lead to reliable convergence to the optimum. Sufficient conditions for probabilistic models were discussed for example in [46]. Nonetheless, probabilistic models that provide provable convergence may often be infeasible despite that the problem is polynomially solvable, as was also shown in our study for two-dimensional spin glasses (see Section VI). On the other hand, handcrafted models restricted strongly according to

the underlying problem structure may often yield otherwise solvable problems intractable, as was shown in Section VI.

Additional studies on the benefits of having an accurate model in EDAs can be found in [74]–[77], which showed that if the probabilistic model matches the structure of the problem, given an infinite population the algorithm will either converge to the optimal solution or the optimal solution will be a stable fixed point of the sampled distribution.

VIII. FUTURE WORK

A direct extension of this work is the analysis of probabilistic models in hBOA on other challenging problems, considering both artificial and real-world problems.

For artificial optimization problems—such as the hierarchical if-and-only-if function [78], [79] and NK landscapes [80]—the primary motivation for studying hBOA models is to gain better understanding of how hBOA works with the goal of improving the algorithm, developing theoretical models, and designing efficiency enhancement techniques. A secondary goal is to gain insights into the reasons why some artificial test problems are easier for hBOA to solve than others.

On the other hand, for real-world problems, the analysis of hBOA models can provide useful information for designing problem-specific variation operators inspired by the learned probabilistic models and focusing the research on challenges identified in important real-world domains.

It should also be interesting to investigate the influence of selection and replacement strategies and their settings on both the performance of hBOA, as well as the structure of probabilistic models discovered by hBOA. One such study has already been published where the structure of probabilistic models and hBOA performance were analyzed for different selection and replacement strategies [10]. Another important step in this direction considers the question of how to modify scoring metrics used for learning Bayesian networks to discover more accurate probabilistic models [81].

Another important topic for future research is the development of automated techniques for effective extraction of relevant information from the analysis of hBOA models and the use of this information for designing efficiency enhancement techniques with the special focus on speeding up model building and learning from experience [2].

IX. CONCLUSION

There are two important reasons for studying probabilistic models obtained by hBOA. First of all, the scalability and efficiency of hBOA as well as the effectiveness of several efficiency enhancement techniques for hBOA crucially depend on the quality, complexity, and stability of the probabilistic models learned by hBOA. Second, the analysis of probabilistic models obtained by hBOA may provide important insights into the structure and difficulty of the underlying problem enabling the design of effective problem-specific efficiency enhancement techniques and variation operators. This paper analyzed the structure, complexity and stability of hBOA probabilistic models on four important classes of test problems: concatenated traps, random additively decomposable problems,

hierarchical traps and two-dimensional Ising spin glasses with periodic boundary conditions.

For concatenated traps, there is a solid body of theory that defines an adequate probabilistic model to ensure that hBOA finds the global optimum scalably and reliably [2], [16], [44]–[46]. The results show that hBOA is able to learn such adequate probabilistic models quickly and that the learned problem decomposition corresponds closely to the underlying problem. The results also show that the models on concatenated traps do not change much from generation to generation, which is important especially for the effectiveness of sporadic and incremental model building.

Also, for random additively decomposable problems, our results show that, despite the varying subproblem difficulty and overlap, hBOA is able to find adequate models, and the models are stable and change only a little during the run.

For hierarchical traps, the results confirm that to solve hierarchical traps, the models must change over time in order to adapt to each level of difficulty as hBOA moves up the hierarchy one level at a time. However, while hBOA works on a specific level of the hierarchy, models change only a little. Just like for concatenated traps and random additively decomposable problems, the models closely correspond to the structure of the underlying problem.

While for two-dimensional Ising spin glasses it is unclear what is an ideal probabilistic model for solving the problem scalably, the probabilistic models obtained by hBOA are shown to correspond closely to the structure of the underlying problem. Furthermore, the results indicate that while it seems to be possible to restrict hBOA models significantly without affecting hBOA performance much, probabilistic models designed by hand may not lead to good scalability despite the robust and scalable performance of hBOA based on automatic model building procedures.

Although the analysis of Bayesian networks with local structures is a challenging task, this paper shows that the analysis of hBOA models is indeed possible and that such an analysis can reveal useful information about the underlying problem. While in the past EDAs were used only as efficient and broadly applicable problem solvers, this paper argues that besides solving the problem, EDAs provide practitioners with information about the underlying problem, which can be used to better understand the problem domain and to design problem-specific efficiency enhancement techniques and operators.

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